

Experiment 1

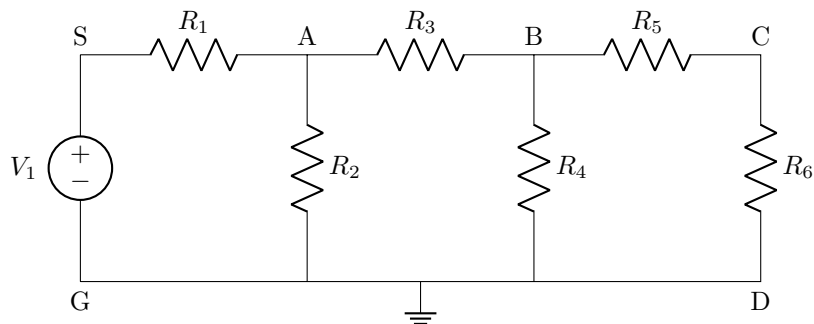
Circuit Laws and Network Theorems

Components/Instruments/Meters

- Breadboard
- Resistances
- DC power supply
- Connecting wires

Part 1 (KCL, KVL)

1. Connect the network as shown below



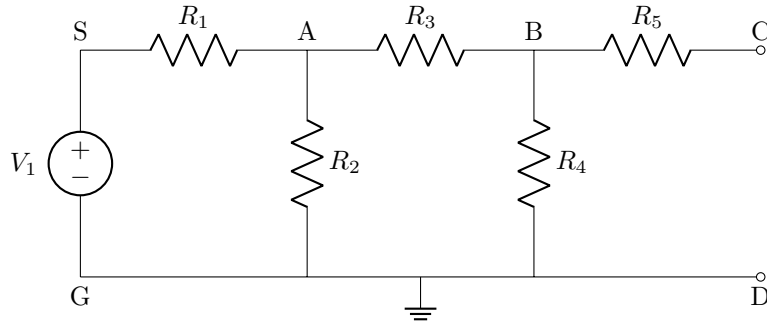
2. Use the following values for the input voltage and resistances

V_1	2V
R_1	100Ω
R_2	$220\ \Omega$
R_3	$100\ \Omega$
R_4	$220\ \Omega$
R_5	$47\ \Omega$
R_6	$150\ \Omega$

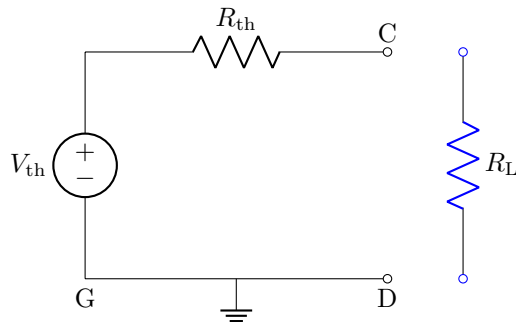
3. Measure voltages across all resistances in the circuit using a multimeter. Calculate the currents expected to flow through these resistors.
4. Verify KCL for node-A and node-B.
5. Verify KVL for all possible loops.

Part 2 (Thevenin's theorem)

1. Connect the same network as Part 1, with R_6 replaced by an open circuit



2. Measure the voltage across C-D.
3. Short C-D and measure the current flowing through it. You can measure the voltage across R_5 and use it to compute the current.
4. Now open C-D and replace V_1 with a short circuit. Measure the resistance across C-D.
5. Obtain the Thevenin's equivalent circuit and compare the measured ' V_{th} ' and ' R_{th} ' with analytically computed values.
6. Is it possible to obtain the Thevenin's equivalent circuit if you skip any of the points 2-4 above?



7. Comment on your observations for different values of R_L . Is the current flowing through R_L same as the one computed using the Thevenin's equivalent circuit?